

Firewall Configuration and Traffic Filtering Report

Objective

Configure and test firewall rules using UFW by enabling the firewall, listing rules, blocking Telnet (23), allowing SSH (22), verifying rules and restoring the original configuration.

Procedure and Commands Executed

Step 1: Update Package Repository

Command:

```
sudo apt update
```

Purpose: Refresh package information.

Step 2: Install UFW

Command:

```
sudo apt install ufw -y
```

Purpose: Install UFW.

Step 3: Verify Installation

Command:

```
ufw version
```

Purpose: Verify successful installation.

Step 4: Check Firewall Status

Command:

```
sudo ufw status
```

Purpose: Confirmed firewall was initially inactive.

Step 5: Enable Firewall

Command:

```
sudo ufw enable
```

Purpose: Enable firewall.

Step 6: View Detailed Status

Command:

```
sudo ufw status verbose
```

Purpose: Verify default firewall policy.

Step 7: List Rules

Command:

```
sudo ufw status numbered
```

Purpose: Display current rules.

Step 8: Block Telnet

Command:

```
sudo ufw deny 23/tcp
```

Purpose: Block insecure Telnet.

Step 9: Verify Rule

Command:

```
sudo ufw status numbered
```

Purpose: Confirm Port 23 block.

Step 10: Allow SSH

Command:

```
sudo ufw allow 22/tcp
```

Purpose: Allow secure SSH.

Step 11: Verify Rules

Command:

```
sudo ufw status numbered
```

Purpose: Confirm SSH rule.

Step 12: Remove Test Rule

Command:

```
sudo ufw delete 1
```

Purpose: Remove temporary rule.

Step 13: Final Verification

Command:

```
sudo ufw status numbered
```

Purpose: Verify final configuration.

Conclusion

The firewall was successfully configured using UFW. The exercise demonstrated how firewall rules filter network traffic, block insecure services, and permit secure remote administration using SSH.